FinSense: Earnings Call Sentiment Analysis using FinBERT

Submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING

DATA SCIENCE

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted to:

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1. Introduction

In modern financial markets, a significant portion of information influencing stock prices is embedded in unstructured textual data such as earnings call transcripts and regulatory filings. Earnings calls, conducted quarterly by company executives (CEO/CFO) with analysts, provide forward-looking insights about company performance, future expectations, risks, and strategic direction. Unlike structured financial data, these textual signals are not always immediately or fully incorporated into stock prices.

This project aims to develop a system that leverages Natural Language Processing (NLP) and domain-specific language models to extract actionable insights from earnings call transcripts. Specifically, the project utilizes a fine-tuned **FinBERT model**, which is pre-trained on financial text, to analyze sentiment embedded in forward-looking statements made by company management.

The core objective is to determine whether forward-looking sentiment derived from earnings calls can be transformed into trading signals that generate abnormal returns (alpha) compared to standard market benchmarks such as the S&P 500.

The proposed system follows a pipeline approach: raw earnings call text is preprocessed and filtered to extract forward-looking statements, which are then passed through the FinBERT model to generate sentiment scores. These scores are transformed into trading signals, which are subsequently evaluated through back-testing strategies.

This project lies at the intersection of **Artificial Intelligence, Financial Analytics, and Quantitative Trading**, and contributes to the growing field of alternative data-driven investment strategies.

2. Feasibility Study

The proposed project is technically and practically feasible due to the availability of pre-trained financial NLP models, open-source datasets, and accessible computational resources.

From a technical perspective, the use of **FinBERT** reduces the complexity of training large-scale models from scratch. Python-based libraries such as HuggingFace Transformers, Pandas, and NumPy provide efficient tools for model implementation and data processing.

From a data perspective, earnings call transcripts and historical stock price data are readily available through public sources and financial platforms, enabling reliable experimentation and validation.

The project is significant as it addresses a key limitation in traditional financial analysis: the underutilization of qualitative textual information. By converting unstructured textual data into quantitative signals, the system enhances decision-making in trading strategies.

Additionally, the project is scalable and can be extended to real-time applications, making it relevant for both academic research and industry applications.

3. Methodology

The project will be developed through the following stages:

1. **Data Collection:**
   Gather earnings call transcripts and corresponding stock price data.
2. **Data Preprocessing:**
   Clean and structure textual data by removing noise, speaker tags, and irrelevant sections.
3. **Forward-Looking Statement Extraction:**
   Identify and extract sentences containing future-oriented information using keyword filtering and rule-based techniques.
4. **Model Fine-Tuning:**
   Fine-tune the FinBERT model on labeled financial text to improve domain-specific sentiment detection.
5. **Sentiment Scoring:**
   Generate sentiment scores for extracted statements based on model predictions.
6. **Signal Engineering:**
   Convert sentiment scores into trading signals using techniques such as absolute sentiment, sentiment change (delta), and surprise factor.
7. **Backtesting:**
   Simulate trading strategies using historical data to evaluate performance.
8. **Performance Evaluation:**
   Analyze results using financial metrics such as alpha, Sharpe ratio, and drawdown.

4. Module & Team Member-wise Distribution of Work

Module Distribution:

Module	Description	Timeline	Assigned Member
Data Collection	Gather transcripts and stock data	Week 1–2	Member 1
Preprocessing	Clean and structure text	Week 3	Member 2
Forward Extraction	Extract forward-looking statements	Week 4	Member 3
Model Fine-tuning	Train FinBERT on dataset	Week 5–6	Member 4
Sentiment Scoring	Generate sentiment outputs	Week 7	Member 4
Signal Generation	Create trading signals	Week 8	Member 5
Backtesting	Implement trading strategy	Week 9–10	Member 5
Evaluation & Report	Analyze results and documentation	Week 11–12	Member 1 & Member 2

Team Member Responsibilities

Member 1: Ankit Kumar Singh - Data & Documentation

Member 2: Ipsita Mondal - Data Processing & Visualization

Member 3: Arindam Ray - NLP Feature Engineering

Member 4: Shubhankar Saket - Model Development

Member 5: Md Rayees Ansari - Quant & Backtesting

5. Innovations in Project

This project introduces several innovations compared to traditional sentiment analysis systems:

- **Forward-Looking Sentiment Extraction:**
  Unlike generic sentiment analysis, this project focuses specifically on future-oriented statements, which are more relevant for predicting stock movements.
- **Domain-Specific Model Adaptation:**
  Fine-tuning FinBERT enhances the model's ability to understand financial language nuances such as uncertainty, guidance, and risk indicators.
- **Delta Sentiment Analysis:**
  The project analyzes changes in sentiment across consecutive earnings calls, capturing shifts in management outlook rather than static sentiment.
- **Integration with Trading Strategy:**
  The system bridges NLP and quantitative finance by converting sentiment into actionable trading signals and validating them through backtesting.
- **Addressing Existing Limitations:**
  Traditional models fail to distinguish between historical and predictive statements. This project specifically filters forward-looking content to improve predictive accuracy.

6. Software and Hardware Requirements

Software Requirements:

- Python (3.x)
- Libraries: Pandas, NumPy, Matplotlib, Scikit-learn
- NLP Framework: HuggingFace Transformers
- Model: FinBERT
- Development Tools: Jupyter Notebook / VS Code

Hardware Requirements:

- System with minimum 8GB RAM
- GPU (optional but recommended for faster fine-tuning)
- Internet access for data collection

7. Bibliography

1. Araci, D. (2019). FinBERT: Financial Sentiment Analysis with Pre-trained Language Models.
2. Devlin, J., et al. (2018). BERT: Pre-training of Deep Bidirectional Transformers.
3. Research papers on earnings call sentiment and stock prediction.
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5. Financial data sources such as Yahoo Finance and Kaggle datasets
6. HuggingFace Transformers documentation